

05-Apr-2026

Agenda:

Stats - 4

- ↳ Quartiles $\rightarrow$ IQR $\rightarrow$ Boundary $\rightarrow$ remove outliers
 - ↳ skewness / kurtosis $\rightarrow$ not that reliable
 - ↳ covariance & correlation
 - ↳ random variable & set
- ↓
hard to interpret

Quartiles $\rightarrow Q_1, Q_2, Q_3 \rightarrow$ 4 parts

IQR $\rightarrow Q_3 - Q_1$

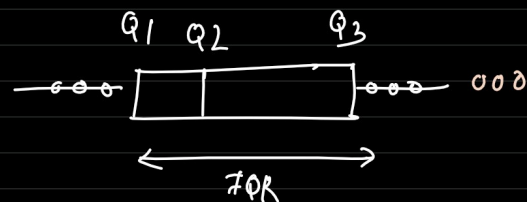
$Q_1 \rightarrow$ 25% tile min -

$Q_2 \rightarrow$ 50% tile max -

$Q_3 \rightarrow$ 75% tile

SD $\downarrow$ mean $\downarrow$ outlier

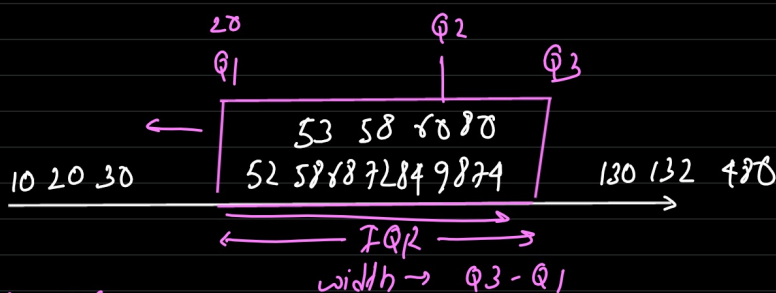
IQR $\rightarrow$ region - width
 $\downarrow$
region where majority of the data lie



The fence method

lower-fence $\rightarrow Q_1 - 1.5 \times IQR$

upper-fence $\rightarrow Q_3 + 1.5 \times IQR$

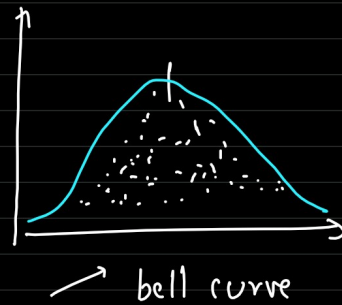


Lower-fence $= Q_1 - IQR \times 1.5$

$$C_{\text{force}} = Q_3 + I_{Q_1} \times 1.5$$

Tulkey (person) $\rightarrow 1.5$

Normal distribution $\rightarrow$ ideal data spread

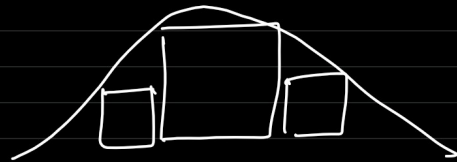


Students,

per month $\rightarrow 25\%$

100 $\rightarrow$ Modest mag $\rightarrow$ 50%.

Good mark - 25%.



Normal distribution $\rightarrow$ $7\sigma \times 1.5 \rightarrow 99.3\%$ all the data
 $\downarrow$
 outside of this region is unusual data

P.1):

4, 8, 12, 15, 18, 21, 24, 27, 30, 33, 38, 42, 47, 51, 94

→ sort data → len() → 15

→ find $Q_1 \rightarrow 25\%$ tile

$Q_1 \rightarrow 25\% \text{ file}$
 $L = 0.25 \times (15-1) + 1 \rightarrow 4.5$
 $4 \rightarrow 15 \rightarrow 16.5$
 $0.5 \rightarrow 0.5 \times 3 = 1.5 \rightarrow$
 5
 Q_1

Q3 → 75% tile

$L \rightarrow 0.75 \times 14 + 1 \rightarrow 11.5 \begin{cases} \nearrow 11 \rightarrow 38 \rightarrow 40 \\ \searrow 0.5 \rightarrow 4 \times 0.5 \rightarrow 2 \rightarrow \uparrow 63 \end{cases}$

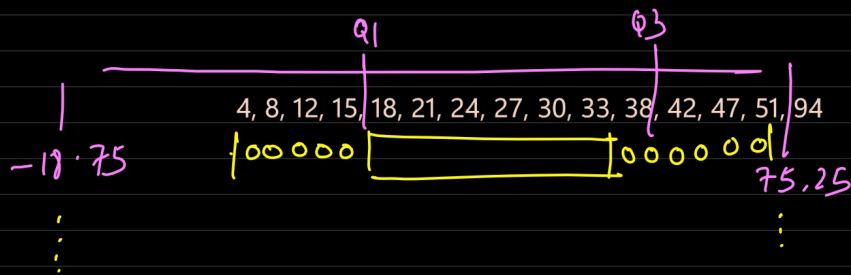
$$TQ_2 \rightarrow Q_3 - Q_1$$

$$\rightarrow 40 - 16.5 \rightarrow 23.5$$

compute the fence:

lower force $\rightarrow 16.5 - (1.5) \times 23.5 \rightarrow 16.5 - 35.25 \rightarrow -18.75$

upper fence $\rightarrow 40 + (1.5) \times 23.5 \rightarrow 40 + 35.25 \rightarrow 75.25$



why we want to remove unusual data?

ML $\rightarrow$ learning

UDS-2.0 $\rightarrow$ Day-1 $\rightarrow$ variable $\rightarrow$ learning
 Day-2 $\rightarrow$ function
 (student) Day-3 $\rightarrow$ function-2 $\rightarrow$
 Day-4 $\rightarrow$ function $\rightarrow$

ML $\rightarrow$

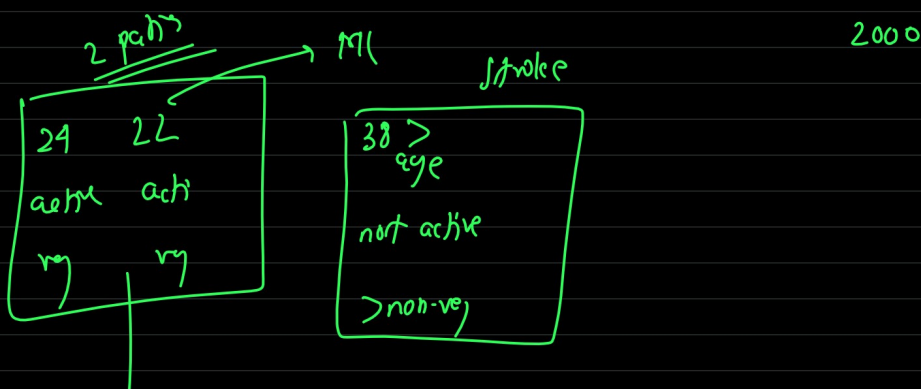
Age	veg/noj-veg	Active	stroke %age
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Age, veg/noj-veg, Active $\rightarrow$ predict stroke %age

clean the unusual data
 $\rightarrow$ for i in data:

ML to learn the pattern

ML $\rightarrow$ formula / pattern (2000 patient)



↓
(create confusion
because it's unusual)

→

Twitter → sample → 300 → 40 data are unus
↓
inputs → $\begin{matrix} X & S & M & L \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$

but if you will remove the outlier then your prediction model will also be biased, isn't it? because you removed some critical data even though it is less likely

ML → Doctor →

ML → weather forecasting → 100%.

1000 → 950 great → desi.
1000 → 1000 okay →
predicted →

$a = [1, 2, 3, 4, 5]$

mask = [True, True, True, True, False]

remove outlier → $a[\text{mask}] \rightarrow [1, 2, 3, 4]$

extract outlier → $\text{mask} = \sim \text{mask}$
 $a[\text{mask}] \rightarrow [5]$

$$arr = [1, 2, 3, 4, 5, 6]$$

$$lower = 2$$

$$upper = 5$$

$$mask = (arr \geq lower) \& (arr \leq upper)$$

$$mask = [f, 1, 1, 1, 1, f] \rightarrow [2, 3, 4, 5] \rightarrow \text{clean data}$$

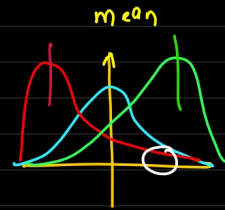
$$\sim mask = [1, f, f, f, f, 1] \rightarrow [1, 6] \rightarrow \text{outliers}$$

Skewness!

Mean $\rightarrow$ center

SD $\rightarrow$ spread

Is data balanced or tilted to the side?

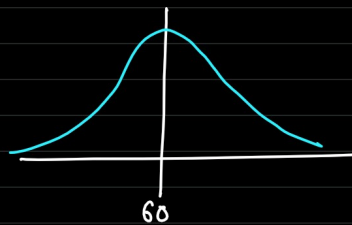


Case 1: Balanced

Data: Marks

$$40, 50, 60, 70, 80 \rightarrow \begin{array}{l} \text{mean} \rightarrow 60 \\ \text{median} \rightarrow 60 \end{array}$$

$$\text{mean} \approx \text{median}$$

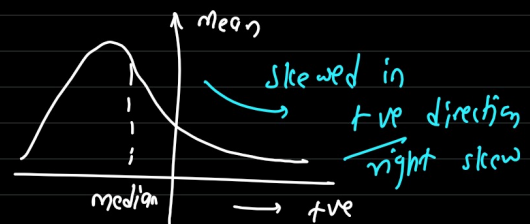
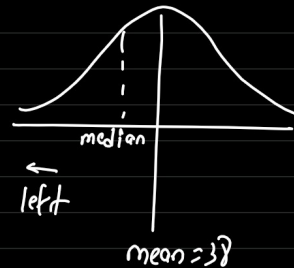


Case 2: Right skew (positive skew)

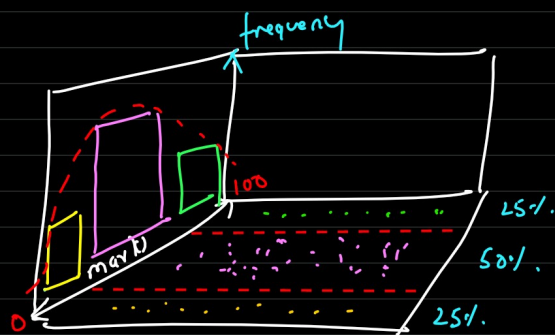
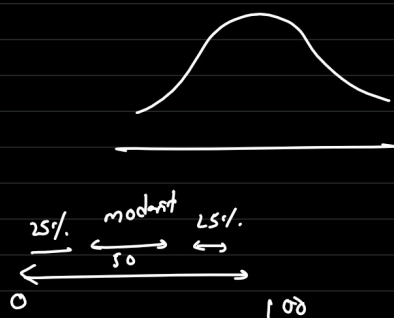
Marks: 10, 20, 30, 40, 90

$$\begin{array}{l} \text{mean} = 38 \\ \text{median} = 30 \end{array}$$

$$30 \approx 38$$



understand the shape of distributions

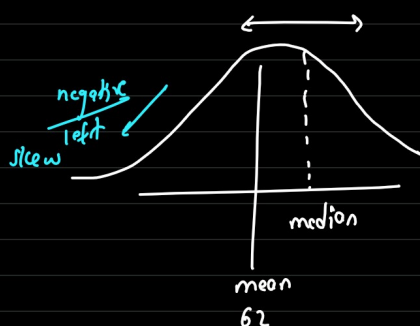


Case 3: left skew (Negative skew)

marks : 10, 60, 70, 80, 90

mean : 62

median : 70



Shape	Relationship (Mean, Median)
Balance	Mean = Median
Tail (Right), positive	Mean > Median
Tail (Left), negative	Mean < Median

positive skew value $\rightarrow$ right skew

Negative skew value $\rightarrow$ left skew

value $\approx 0 \rightarrow$ symmetric data

skewness formula: $\frac{\sum (x_i - \mu)^3}{n \cdot \sigma^3} \rightarrow +ve$

$x_i \rightarrow$ individual data point

$\mu \rightarrow$ Mean of the dataset

n : number of data point

σ : standard deviation

col 1 col 2 col 3 col 4

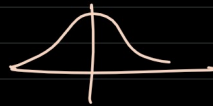
kurtosis

problem:

Are extreme value common?

$\rightarrow$ Balanced

0



$\rightarrow$ Few big outliers

High

$\rightarrow$ Few small outliers

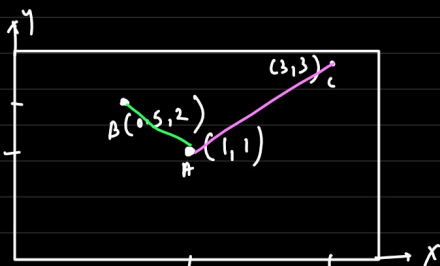
High

$\rightarrow$ flat distribution

Low



$$\frac{\sum (x_i - \mu)^4}{n \cdot \sigma^4}$$

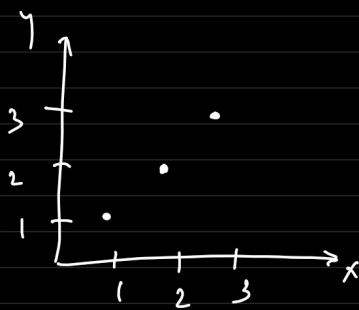


$$A \rightarrow B \rightarrow \sqrt{(0.5)^2 + (1)^2} \rightarrow \sqrt{0.25 + 1} \rightarrow \sqrt{1.25} \rightarrow 1.118$$

$$A \rightarrow C \rightarrow \sqrt{(2)^2 + (2)^2} \rightarrow \sqrt{8} \rightarrow 2.83$$

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \rightarrow$$

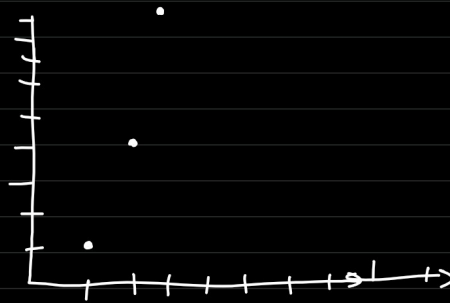
* Square $\rightarrow$ helps us remove the directional sign from the value



A $\rightarrow$ (1,1)

B $\rightarrow$ (2,2)

C $\rightarrow$ (3,3)



	x	y ²
A $\rightarrow$	1	1
B	2	4
C	3	9

≈ 0

Normal

> 0

outlier value is extreme

< 0

Few outliers

openai → chatgpt
→ "key" → secret → "

openai install

llm = open("key", "gpt-5")

resp = llm.invoke("explain llm & provide the answer in dict")
↓

python → llm

workflow :

input: "my id is 123"

↓
llm → extract user_id → 123

~~database~~ python (123) →

↓
yes, no

↓
yes → "Awhichis" → "upl.d"